# A unified data preprocessing framework to address inconsistencies in comparative plastid genome studies

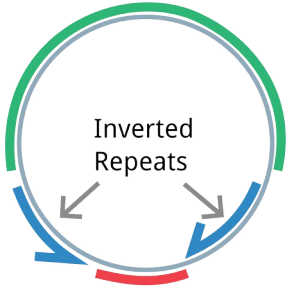
A. Emirsaliev [1, 2, 3] *, E. Salina [1, 2], I. Mitrofanova [3], D. Afonnikov [1, 2]

[1] Institute of Cytology and Genetics, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russia;
[2] Kurchatov Genomics Center, Institute of Cytology and Genetics, Siberian Branch of the Russian Academy of Sciences, Novosibirsk, Russia;
[3] N.V. Tsitsin Main Botanical Garden of Russian Academy of Science, Moscow, Russia
* *emirsaleh@bionet.nsc.ru*

# The importance of plastid genomes in plant phylogeny reconstruction

**Plastid genome characteristics**

- circularized DNA
- ~150 000 bp
- ~80 protein-coding genes
- two inverted repeats of ~20 000 bp
- variable intergenic spacers -> ~20 barcodes




Plastid data can be used to resolve phylogeny at different scale – both for close relatives and within entire phylum.

JOURNAL ARTICLE

**Phylogeny and evolution of Asparagaceae subfamily Nolinoideae: new insights from plastid phylogenomics**

Yunheng Ji, Jacob B Landis, Jin Yang, Shuying Wang, Nian Zhou, Yan Luo, Haiyang Liu

*Annals of Botany*, Volume 131, Issue 2, 1 February 2023, Pages 301–312, https://doi.org/10.1093/aob/mcac144

**Published:** 26 November 2022 **Article history**

**Comparative analysis of plastid genomes within the Campanulaceae and phylogenetic implications**

Chun-Jiao Li, Ruo-Nan Wang, De-Zhu Li

Published: May 14, 2020 • https://doi.org/10.1371/journal.pone.0233167

**Plastid phylogenomic insights into relationships of all flowering plant families**

Hong-Tao Li, Yang Luo, Lu Gan, Peng-Fei Ma, Lian-Ming Gao, Jun-Bo Yang, Jie Cai, Matthew A. Gitzendanner, Peter W. Fritsch, Ting Zhang, Jian-Jun Jin, Chun-Xia Zeng, Hong Wang, Wen-Bin Yu, Rong Zhang, Michelle van der Bank, Richard G. Olmstead, Peter M. Hollingsworth, Mark W. Chase, Douglas E. Soltis, Pamela S. Soltis, Ting-Shuang Yi & De-Zhu Li

*BMC Biology* 19, Article number: 232 (2021) | Cite this article

Front. Plant Sci., 11 April 2022
Sec. Plant Systematics and Evolution
Volume 13 - 2022 | https://doi.org/10.3389/fpls.2022.808156

This article is part of the Research Topic
Phylogenomic Discordance in Plant Systematics
View all 15 articles

**Comparative Analyses of 3,654 Plastid Genomes Unravel Insights Into Evolutionary Dynamics and Phylogenetic Discordance of Green Plants**

Ting Yang, Sunil Kumar Sahu, Lingxiao Yang, Yang Liu, Weixue Mu, Xin Liu, Mikael Lenz Strube, Huan Liu, Bojian Zhong

Front. Plant Sci., 01 February 2018
Sec. Evolutionary and Population Genetics
Volume 9 - 2018 | https://doi.org/10.3389/fpls.2018.00082

**Plastid Genome Comparative and Phylogenetic Analyses of the Key Genera in Fagaceae: Highlighting the Effect of Codon Composition Bias in Phylogenetic Inference**

Yanci Yang, Juan Zhu, Li Feng, Tao Zhou, Guoqing Bai, Jia Yang, Guifang Zhao

**A plastid phylogeny of the fern genus *Arachniodes* (Dryopteridaceae)**

Ngan Thi Lu, Atsushi Ebihara, Hai He, Liang Zhang, Xin-Mao Zhou, Ralf Knapp, Peris Kamau, David Lorence, Xin-Fen Gao, Li-Bing Zhang

Show more

+ Add to Mendeley Share Cite

https://doi.org/10.1016/j.ympev.2018.12.013 Get rights and content

**Molecular Phylogeny of the *Lactuca* Alliance (Cichorieae Subtribe Lactucinae, Asteraceae) with Focus on Their Chinese Centre of Diversity Detects Potential Events of Reticulation and Chloroplast Capture**

Ze-Huan Wang, Hua Peng, Norbert Kilian

Published: December 20, 2013 • https://doi.org/10.1371/journal.pone.0082692

JOURNAL ARTICLE

**Phylogeny and evolution of Asparagaceae subfamily Nolinoideae: new insights from plastid phylogenomics**

Yunheng Ji, Jacob B Landis, Jin Yang, Shuying Wang, Nian Zhou, Yan Luo, Haiyang Liu

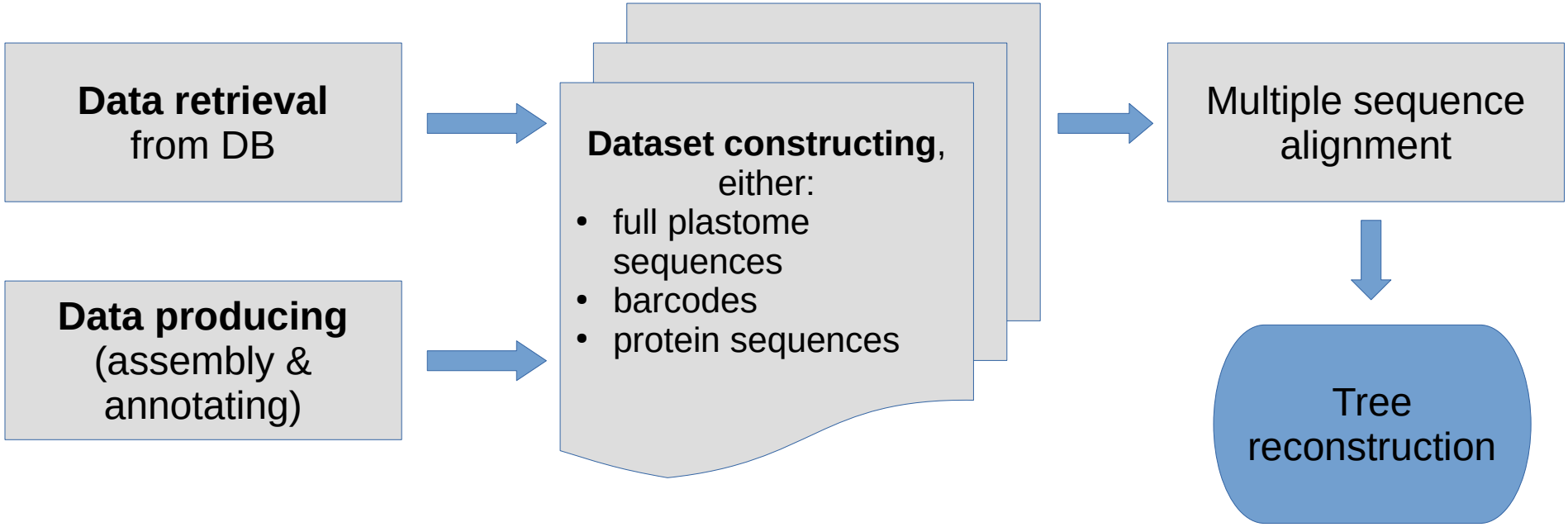
*Annals of Botany*, Volume 131, Issue 2, 1 February 2023, Pages 301–312, https://doi.org/10.1093/aob/mcac144

**Published:** 26 November 2022 **Article history**

Reasons for a wide use of plastid data in phylogeny

- high copy number in cell
- conserved gene order and content
- relatively small size
- mostly maternal inheritance
- widely used barcode sequences
- low per sample analysis costs
- a huge corpus of data collected

# Generalized workflow in plastid phylogeny study



# Why do we need plastome data standardization?

**Data inconsistency sources**

- structural inconsistencies
- improper annotation
- wrong metadata
- record duplications
- etc.

**Shortcommings**

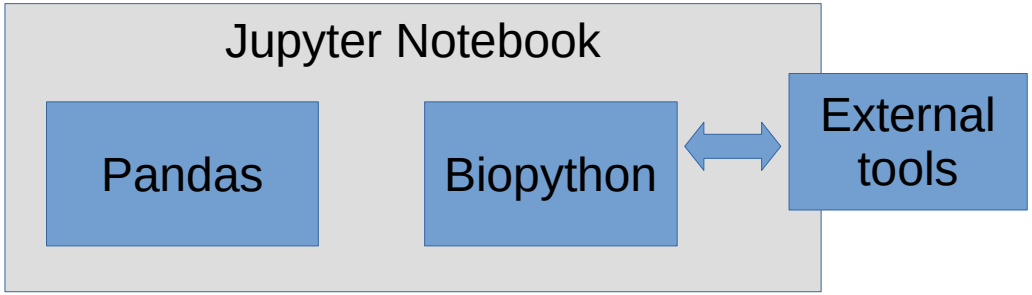
- challenges in database retrieval
- gaps in dataset
- alignment quality decrease
- multiple names for the same entity
- etc.

# A framework addressing data quality and comparability issues

Plastid genome structure is considered to be very conserved. Data quality might greatly affect analysis, especially within close-related species.

Among Asteraceae accessions, sequence standardization increased alignment length to 30%, and the number of aligned genes for two times.*

## Technologies used




* DOI: 10.3390/plants10071360

## The framework might led to data quality improvement at the following stages:

1) Data querying and retrieval from Nucleotide Database of NCBI/EMBL-EBI/DDBJ via Entrez API.
2) Records deduplication.
3) Assessing the coverage of annotation.
4) Validating the linearized representation of plastome sequences.
5) Defining the records to be kept.
6) Properly ordering and orienting genome regions.
7) Filling missing but required fields in the annotation.
8) Extracting data for analysis based on the purpose.

# Challenges in data retrieval

Naming patterns that co-occure

**FASTA header examples**

>OL891805.1 UNVERIFIED: Soroseris gillii **chloroplast sequence**

>OM859427.1 UNVERIFIED: Martinella insignis voucher A. R. Zuntini 151 (SPF) **chloroplast, partial genome**

>OR771013.1 UNVERIFIED: Cuscuta japonica **plastid sequence**

>NC_051973.1 Stebbinsia umbrella voucher ZJW3542 **chloroplast, complete genome**

>NC_082051.1 Gaillardia pulchella **chloroplast, complete sequence**

**Solution**: complex queries

```
target = f’(“chloroplast sequence” OR “plastome” OR “plastid, complete genome”
OR “plastid sequence” OR “chloroplast, complete sequence”)’
query = f’{target} AND “Asteraceae”[Organism]’
```

# Challenges in data retrieval

Query responses might contain off-target data

**Out-of-target records in query response**

>AF097517.1 Helianthus annuus ribulose-1,5-bisphosphate carboxylase/oxygenase (rbcL) gene, plastid gene encoding plastid protein, complete cds

>OQ826672.1 Saussurea baicalensis voucher KHU22-12-01 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

>EU391981.1 Tragopogon albinervis psbA-trnH intergenic spacer, partial sequence; chloroplast

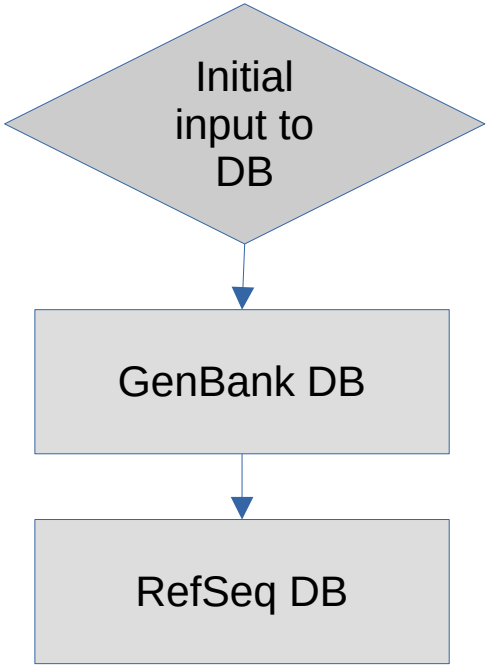
>EU391974.1 Tragopogon marginifolius PsbA (psbA) gene, partial cds; and psbA-trnH intergenic spacer, partial sequence; chloroplast

**Solution**: more complex queries

```
target = f’(“chloroplast sequence” OR “plastome” OR “plastid, complete genome” OR “plastid sequence”)’
nots = ‘(“spacer” OR “gene” OR “cds” OR “partial” OR “ITS” OR “genes” OR “CDS”)’
query = f’{target} AND “Asteraceae”[Organism] NOT {nots}’
```

# Record deduplication

A typical dataflow in record depositing




The references to the initial deposition are noted in database record metadata of the RefSeq accessions:

```
COMMENT     PROVISIONAL REFSEQ: This record has not yet been subject to final
            NCBI review. The reference sequence was derived from AP007232.
            COMPLETENESS: full length.
```

```
COMMENT     PROVISIONAL REFSEQ: This record has not yet been subject to final
            NCBI review. The reference sequence is identical to OR124735.
            COMPLETENESS: full length.
```

The map of accession identifiers

| Organism | RefSeq ID | INSDC ID |
|---|---|---|
| *Crepis rigescens* | NC_061360.1 | OM320794.1 |
| *Cichorium intybus* | NC_043842.1 | MK569377.1 |
| *Crepidiastrum denticulatum* | NC_042149.1 | MK622902.1 |
| *Crepidiastrum lanceolatum* | NC_046512.1 | MK358413.1 |
| *Lactuca sativa* | NC_007578.1 | AP007232.1 |

# Inconsistent feature naming

## A case of genes

**Alternative names of the same genes among plastid genomes of different Cichorieae accessions, $i_{max}$=15**

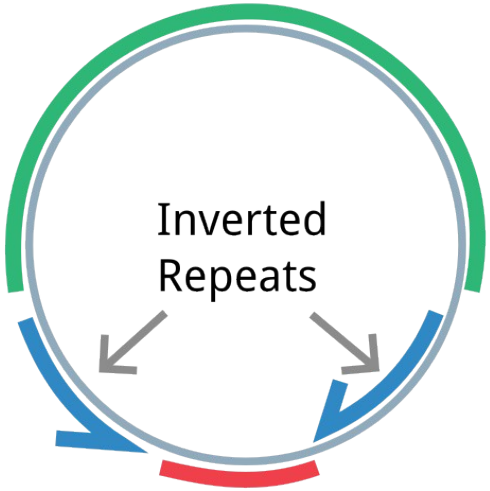
- trnT-UGU, trnT, trnT-TGT, trnT-GGT, trnT-GGU, trnT-fragment
- trnI-GAU, trnE-UUC, trnE, trnI, trnII-GAU
- pafI, ycf3, I9990_pgp067
- pafII, ycf4, I9990_pgp057
- clpP1, clpP, HRV17_pgp042, I9896_pgp042, I9987_pgp041, I9990_pgp042, I9991_pgp042, I9992_pgp042
- ndhB, HRV17_pgp004, HRV17_pgp022, I9819_pgp004, I9819_pgp022, I9896_pgp004, I9896_pgp022, I9987_pgp004, I9987_pgp021, I9990_pgp004, I9990_pgp022, I9991_pgp004, I9991_pgp022, I9992_pgp004, I9992_pgp022
- rps12, HRV17_pgp043, I9896_pgp043, trans splicing 5'-rps12 and 3'-rps12 in IRA, trans splicing 5'-rps12 and 3'-rps12 in IRB
- rrn16, rrn16S
- etc.

15 names

# Inconsistent feature naming

## A case of genomic regions

A typical structure of Embryophyta plastid genome




Inverted repeats

**Feature type:**

- repeat_region
- misc_feature

**Feature names:**

- inverted repeat A (IRA)
- inverted repeat A region(IRa)
- inverted repeat A region; IRA repeat region; Inverted repeat A region
- inverted repeat A
- inverted repeat (IR)
- inverted repeat region (IR)
- IRA

Single copy regions

**Feature type:**

- misc_feature

**Feature names:**

- large single copy region(LSC)
- large single copy region; LSC
- large single copy (LSC)

# Missing annotations

## The completeness of annotations

A. Inverted repeats

| Accession | Organism | LSC | SSC | IR |
|---|---|---|---|---|
| NC_031815 | Taraxacum obtusifrons | 0 | 0 | 2 |
| NC_057108 | Ixeris repens | 1 | 1 | 2 |
| NC_046513 | Crepidiastrum sonchifolium | 0 | 0 | 2 |
| NC_042149 | Crepidiastrum denticulatum | 0 | 0 | 0 |

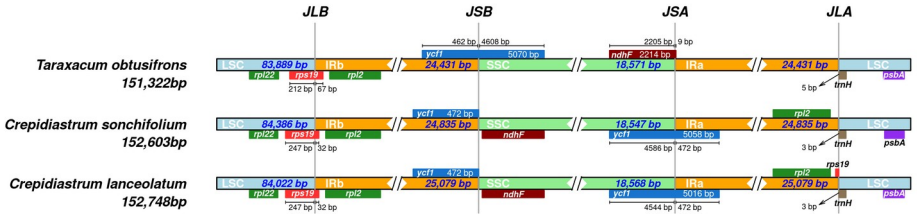
B. Features of type gene

| Accession | rbcL | accD | psbH | pbf1 |
|---|---|---|---|---|
| NC_062824.1 | 1 | 1 | 1 | 0 |
| NC_061360.1 | 1 | 1 | 1 | 0 |
| NC_059769.1 | 1 | 0 | 1 | 0 |
| NC_059768.1 | 1 | 0 | 1 | 0 |
| NC_057108.1 | 1 | 1 | 1 | 0 |

# Common issues with genome regions

- IRs differ by lengths
- IRs are not annotated
- wrong feature type was assigned to IR
- inconsistent feature naming
- sequence start position was shifted
- LSC and SSC are not annotated
- accessions differ in region orientation
- etc.

IR junction sites are widely used in phylogenetic studies

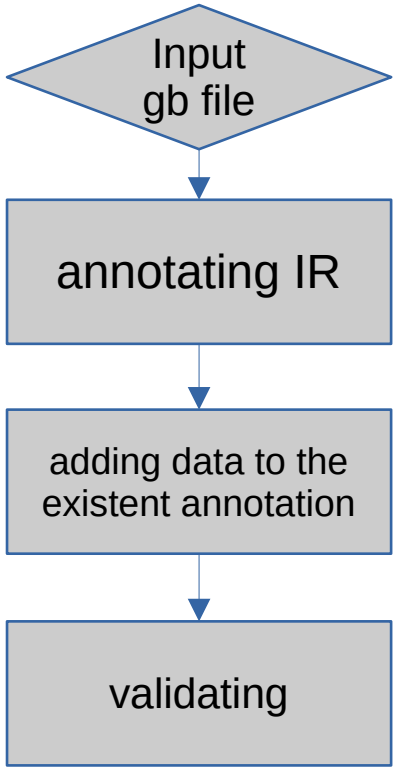



Summary of issues within Crepidinae dataset:

- 15 accessions lacks IR annotations
- 5 have IR that were annotated as “repeat_region” with “note” field left blank
- 10 have no LSC and SSC regions annotated
- dataset is inconsistent in SSC orientation
- truncated genes spanning junction sites are often missed in annotation

# Fixing annotation

IR annotation issue resolving workflow



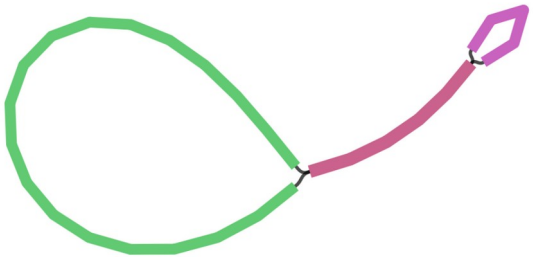

Software options for programmatic IR annotation on local machine

| | **PGA.pl** | **Chloë** | **NOVOWrap** | **CpDNA_standardization** |
|---|---|---|---|---|
| annotating IR | x | x | x | |
| rotating | | x | x | |
| orienting | | | x | x |
| validating | | | x | |

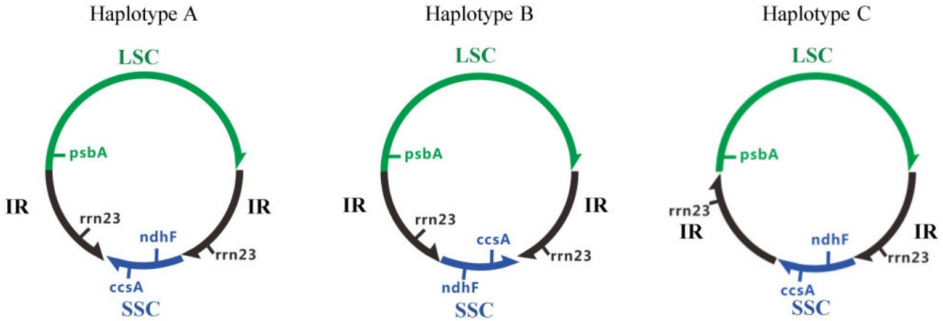
Some accessions has feature hierarchical order mixed, that may brake downstream analysis, e.g. feature of type “gene” mentioned after corresponding feature “CDS”. It should also be fixed.

# Standardizing structure for depositing

The graph of *Seseli lehmannii* plastid genome assembly

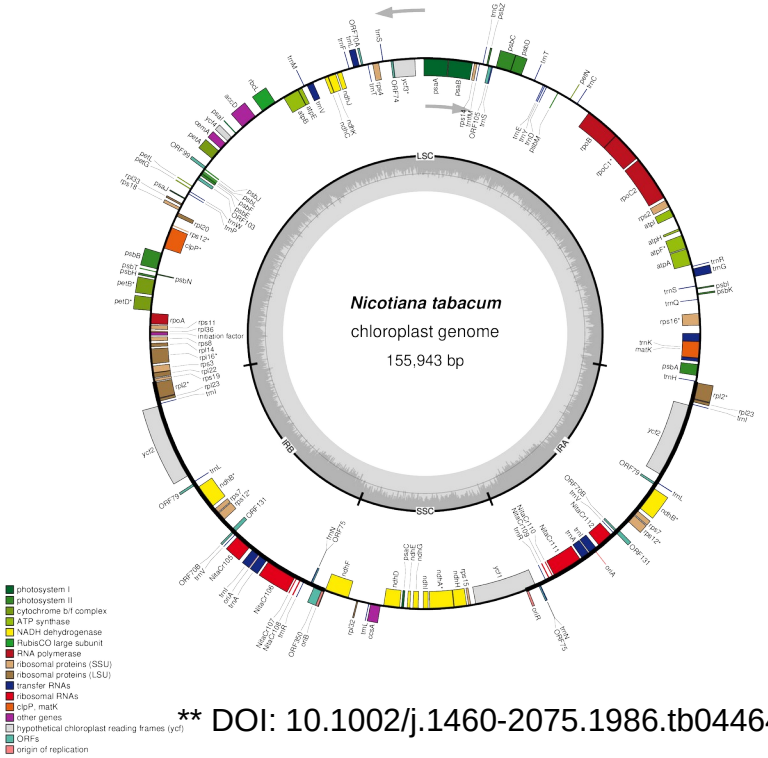


The most frequent isoforms of plastid genome circular DNA *




* DOI: 10.1093/gbe/evz256

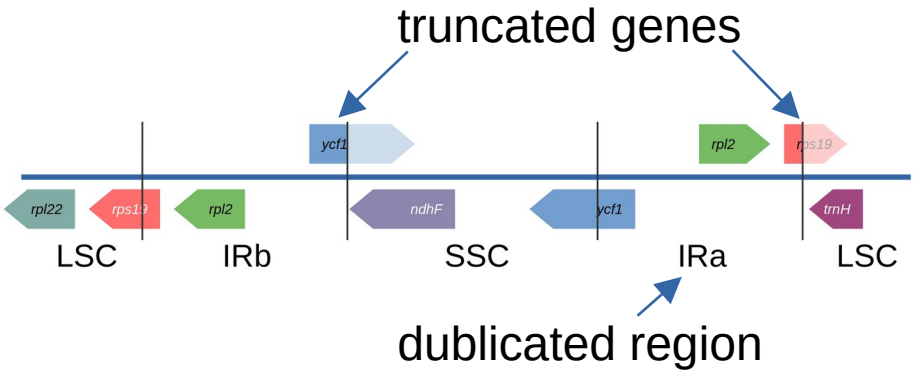
The plot of annotated plastid genome of Nicotiana tabacum (RefSeq ID: NC_001879.2)**

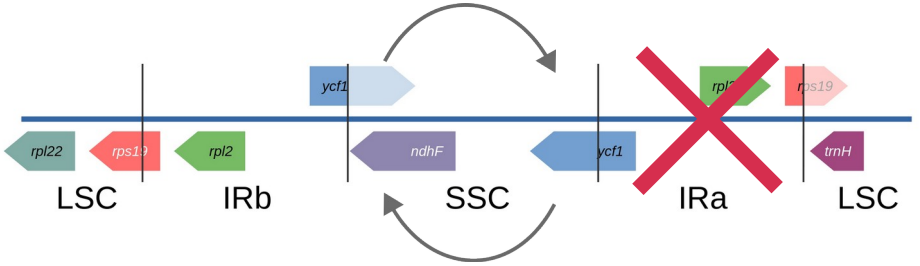


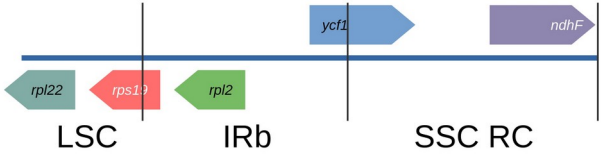

** DOI: 10.1002/j.1460-2075.1986.tb04464.x

# Standardizing structure for analysis

- Excluding the second repeat
- Orienting regions so that genes span region junctions



# Fixing metadata

Programmatic annotators and genome browsers often omits some fields

**Solution**: filling gaps programmatically using Biopython

```
# define custom lineage
crepis_taxonomy = ["Eukaryota", "Viridiplantae", "Streptophyta", "Embryophyta",
"Tracheophyta", "Spermatophyta", "Magnoliopsida", "eudicotyledons", "Gunneridae",
"Pentapetalae", "asterids", "campanulids", "Asterales", "Asteraceae", "Cichorioideae",
"Cichorieae", "Crepidinae", "Crepis"]
# add lineage
record.annotations['taxonomy'] = crepis_taxonomy

# add data to header line
record.annotations["molecule_type"] = "DNA"
record.annotations["data_file_division"] = "PLN"
record.annotations["topology"] = "circular"
```